

Data Analytics Final Project Guidelines

Project Objective

The main objective of this project is to **analyse hydrological and agricultural data from the Cauvery River basin** to understand water availability, reservoir performance, and irrigation water stress between Karnataka and Tamil Nadu.

Specific Objectives:

- To analyse reservoir storage, river discharge, and water levels across Karnataka and Tamil Nadu.
- To quantify agricultural water deficits in the Cauvery delta districts.
- To compare upstream (Karnataka) and downstream (Tamil Nadu) water availability.
- To evaluate the growing dependence on groundwater due to insufficient canal supply.
- To support data-driven reservoir release planning for better agricultural water management.

Project Scope:

This project covers the complete data analytics workflow for the Cauvery River basin datasets, which includes:

1. **Data Loading and Initial Overview**
 - Loading four datasets (Water Level, River Discharge, Reservoirs, and Agricultural Water Demand)
 - Examining structure, data types, and summary statistics
2. **Data Pre-processing**
 - Handling missing values and duplicates
 - Treating outliers in key numerical columns
 - Creating derived columns such as Season, Basin Zone (Upstream/Downstream), Net Flow, Year, and Month
 - Standardizing and transforming the data for analysis
3. **Exploratory Data Analysis (EDA)**
 - Univariate Analysis
 - Bivariate Analysis
 - Multivariate Analysis
4. **Data Visualization**
 - Creating meaningful charts using Matplotlib and Seaborn (more than 10 visualizations)
5. **Insight Generation**
 - Identifying key patterns related to water availability, reservoir performance, and agricultural water deficits
 - Providing recommendations based on the analysis.

Project Domains

- **Environmental:** Hydrology & Agriculture Analytics

Dataset Requirements

- As per the project guidelines, the dataset should contain a minimum of 500 records and at least 10 features.
- This project uses four datasets related to the Cauvery River basin (Water Level, River Discharge, Reservoirs, and Agricultural Water Demand) for the period 2021–2026. All datasets individually and collectively satisfy the minimum requirements of 500+ records and 10+ features, containing a mix of numerical and categorical columns suitable for detailed exploratory data analysis.

Data Sources

- Government Portals / (India Water Resources Information System (India-WRIS) / Central Water Commission (CWC) / National Water Data Portal (NWDP)).
- <https://nwdp.nwic.gov.in/dataset/river-water-level-telemetry-hourly-central-water-commission-cwc>,
- <https://nwdp.nwic.gov.in/dataset/river-discharge-manual-dailly-central-water-commission-cwc>,
- https://www.tnagrisnet.tn.gov.in/home/reservoir/%7B%7Bdata.apply_link%7D%7D
- <https://nwdp.nwic.gov.in/>

Submission Requirements

1. Original Dataset Source

The raw datasets used for this project are provided in CSV/Excel format as obtained from the respective data sources. The datasets contain information related to Cauvery water levels, river discharge, reservoir conditions, and agricultural water demand and supply.

2. Jupyter/Google Colab Notebook

The completed. ipynb notebook contains the end-to-end data analytics process, including:

- Data loading and initial exploration
- Data cleaning and pre-processing
- Handling missing values and duplicate records
- Data type correction and formatting
- Outlier detection and treatment
- Creation of derived columns
- Univariate, bivariate, and multivariate analysis

- Groupby, aggregation, pivot table, and correlation analysis
- Statistical summaries
- More than 10 meaningful visualizations using Python libraries such as Matplotlib and Seaborn
- Markdown explanations for the methodology, analysis, and interpretation of results
- Key findings related to water availability, reservoir conditions, agricultural demand, water supply, and water deficit.

3. Key Insights

The project identifies key patterns in Cauvery water availability and agricultural water requirements across the study period. Important findings include:

- Seasonal variations in agricultural water demand and water deficit.
- Differences in water availability between upstream and downstream regions.
- Reservoir inflow, outflow, and storage patterns.
- Relationship between canal water supply and groundwater extraction.
- District-wise and season-wise differences in agricultural water deficits.

4. Final Project Output

The project provides a data-driven understanding of Cauvery water availability and agricultural water requirements using Python-based Exploratory Data Analysis. The findings can help understand seasonal water shortages, reservoir conditions, and the dependence on different water sources.

Data Analytics Final Project Rubric

TASK 1: Problem Definition & Dataset Selection

Problem Statement

The objective of this project is to analyze Cauvery water availability, reservoir conditions, river discharge, and agricultural water demand across different regions and seasons. The analysis aims to identify water availability patterns, agricultural water deficits, and the dependence on canal water and groundwater.

Dataset Selection

The project uses datasets related to:

- Water level monitoring
- River discharge
- Reservoir inflow, outflow, and storage
- Agricultural water demand and supply
- District and seasonal information

The datasets cover the study period from **2021 to 2026** and contain both numerical and categorical features.

TASK 2: Data Cleaning & Pre-processing

The datasets were cleaned and prepared before performing Exploratory Data Analysis.

Data Pre-processing Steps

- Loaded the datasets using Pandas.
- Checked the number of rows and columns.
- Examined data types using info().
- Checked missing values.
- Identified and handled duplicate records.
- Corrected inappropriate data types.
- Standardized column names and formats.
- Checked for inconsistent categorical values.
- Detected outliers using statistical methods and box plots.
- Applied appropriate outlier treatment where required.
- Created derived columns such as:
 - Discharge Category
 - Water Deficit
 - Deficit Percentage
 - Canal Water Dependency
 - Groundwater Dependency
- Filtered and aggregated data for district-wise, seasonal, and yearly analysis.
- Combined relevant datasets where required for further analysis.

TASK 3: Exploratory Data Analysis (EDA) & Visualizations

The cleaned datasets were analyzed using **Univariate, Bivariate, and Multivariate Analysis**.

Univariate Analysis

Individual variables were analyzed to understand their distribution and statistical characteristics.

Examples:

- Water level distribution
- River discharge distribution
- Reservoir inflow and outflow
- Agricultural water demand
- Water deficit percentage

Bivariate Analysis

Relationships between two variables were analyzed.

Examples:

- Water demand vs water supplied
- Canal water vs groundwater extraction
- Reservoir inflow vs outflow
- District vs water deficit

Multivariate Analysis

Multiple variables were analyzed together to identify more complex patterns.

Examples:

- District-wise water deficit across different seasons
- Demand, supply, and groundwater usage across years
- Reservoir conditions and water availability
- Seasonal agricultural water requirements

Statistical Analysis

The project includes:

- Mean
- Median
- Standard deviation
- Variance
- Minimum and maximum values
- Skewness
- Correlation analysis
- Groupby analysis
- Pivot tables
- Aggregated summaries

Visualizations

More than **10 visualizations** were created using Matplotlib and Seaborn to identify patterns and trends.

Examples include:

1. Water Level Distribution
2. Water Level by State/Region
3. River Discharge Trend
4. Discharge Category Distribution
5. Reservoir Inflow Analysis
6. Reservoir Outflow Analysis
7. Reservoir Storage Percentage
8. Agricultural Water Demand vs Supply
9. Water Deficit by District

10. Water Deficit by Season
11. Canal Water vs Groundwater
12. Correlation Heatmap
13. Outlier Analysis using Box Plots
14. Year-wise Water Demand and Supply Analysis

Each visualization contains an appropriate title, axis labels, legend where required, and an interpretation of the result.

TASK 4: Documentation, Insights & Presentation

Documentation

The notebook is organized into different sections covering:

- Introduction
- Problem Definition
- Dataset Description
- Data Loading
- Data Understanding
- Data Cleaning
- Data Pre-processing
- Exploratory Data Analysis
- Statistical Analysis
- Visualizations
- Key Insights
- Conclusion

Markdown cells are used throughout the notebook to explain the purpose of each analysis, the methodology followed, and the interpretation of the results.

Key Insights

The analysis identified several important findings:

1. **Seasonal variation:** Agricultural water demand and water deficits vary across different seasons.
2. **District-level variation:** Water deficit differs across the Cauvery delta districts, indicating that water requirements are not uniform across regions.
3. **Reservoir patterns:** Reservoir inflow, outflow, and storage levels show variations over time, reflecting changes in water availability.
4. **Water-source dependency:** Agricultural water requirements are supported by both canal water and groundwater, with the level of dependence varying by season and region.
5. **Demand and deficit:** Differences between agricultural water demand and water supplied indicate periods where additional water resources may be required.

Conclusion

The project provides a data-driven analysis of Cauvery water availability and agricultural water requirements from 2021 to 2026. Through data cleaning, statistical analysis, EDA, and visualization, the study identifies patterns in water levels, discharge, reservoir conditions, agricultural demand, supply, and water deficits. The analysis demonstrates how data analytics can be used to understand water-resource patterns and support further planning and monitoring.